Uniform Rational Approximations to an Elliptic Logarithm

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Abstract

For the elliptic curve $y^2 = x^3 - Dx$, we give an explicit rational polynomial approximation to the formal logarithm, with a local error bound uniform in the truncation length. The estimate applies at every odd unramified prime and does not require the prime to exceed the truncation order. The same polynomials are positive on the real interval $[0, 1]$, which proves nonvanishing of the associated logarithms at every nonzero double on the specified curve over $\mathbb{Q}(\sqrt{2})$. Integer coefficient formulas and norm bounds provide exact finite specifications for arbitrarily long truncations. All statements in this paper are unconditional.

1 Uniform logarithm truncations of arbitrary order

Let $E : y^2 = x^3 - Dx$ over a field of characteristic zero, with $D \neq 0$. Write $f = 1/x$ and $z = -x/y$ near the identity. The equation gives

$$f = z^2(1 - Df^2).$$

Define rational numbers

$$c_0 = 1, \qquad c_1 = -\frac{2}{5}, \qquad c_j = c_{j-1}\frac{4j-5}{4j+1} \quad (j \geq 2),$$

and put

$$Q_J(T) = \sum_{j=0}^{J-1} c_j T^j, \qquad L_J(Q) = z(Q)Q_J(Df(Q)^2) \quad (J \geq 1).$$

Lemma 1.1. *The formal series*

$$L(Q) = z(Q)\sum_{j\geq 0} c_j (Df(Q)^2)^j$$

is the normalized formal logarithm of the displayed equation. For every odd prime p and $j \geq 1$,

$$v_p(c_j) \geq -\lfloor \log_p(4j+1) \rfloor.$$

Proof. The invariant differential is $\omega = dx/(2y) = z\,df/(2f)$. Put $T = Df^2$ and $S(T) = \sum c_j T^j$. From $z^2 = f/(1-T)$, differentiation shows that $d(zS(T)) = \omega$ is equivalent to

$$4T(1-T)S'(T) + (1+T)S(T) = 1 - T.$$

The constant coefficient is $c_0 = 1$, the linear coefficient is $5c_1 = -2$, and every subsequent coefficient is precisely the stated recurrence. The resulting series is $z + O(z^5)$ and hence is the normalized formal logarithm.

For $j \geq 1$ its coefficient is exactly

$$c_j = 2 \prod_{i=1}^{j} \frac{4i-5}{4i+1}.$$

For an odd prime power p^r, each of the two arithmetic progressions in this quotient has length j and invertible step four. The counts of terms divisible by p^r therefore differ by at most one. No term has absolute value exceeding $4j+1$. Summing the possible loss of one valuation over the prime powers at most $4j+1$ proves the claimed lower bound. □

Theorem 1.2. *Let $F/\mathbb{Q}_p$ be unramified, where p is odd, normalize $v(p) = 1$, and suppose D is integral. For any nonzero point Q in the formal group, so that $v(z(Q)) \geq 1$, and every integer $J \geq 1$,*

$$v(L_J(Q)) = v(z(Q)),$$
$$v(L(Q) - L_J(Q)) \geq v(z(Q)) + 3J.$$

In particular, the conclusion is uniform in J and does not require $p > J$ or $p > 4J$.

Proof. The relation between f and z gives $v(f(Q)) = 2v(z(Q))$. For every $j \geq 1$ and odd p one has

$$\lfloor \log_p(4j+1) \rfloor \leq j.$$

Indeed $3^{j+1} > 4j+1$ for $j \geq 1$, by induction. Consequently Lemma 1.1 gives

$$\begin{aligned} v(z(Q)c_j(Df(Q)^2)^j) &\geq v(z(Q)) + 4jv(z(Q)) - \lfloor \log_p(4j+1) \rfloor \\ &\geq v(z(Q)) + 3j. \end{aligned}$$

All nonconstant terms are strictly smaller in absolute value than the leading term $z(Q)$, proving the first equality. The same estimates show that the infinite series converges, and that its tail beginning at $j = J$ has the stated valuation. □

Corollary 1.3. *In the setting of Theorem 1.2, suppose $Q = [n]P$ and $R = [m]P$ are nonzero formal points, where m, n are nonzero integers. For truncation lengths $J_1, J_2 \geq 1$, put $k = 3\min(J_1, J_2)$. Then*

$$v\left(\frac{L_{J_1}(Q)}{L_{J_2}(R)} - \frac{n}{m}\right) \geq v(n/m) + k.$$

If n/m is integral, the right side is at least k.

Proof. Formal additivity applied to $[m]Q = [n]R$ gives $L(Q)/L(R) = n/m$. The leading-term estimate in the theorem makes both logarithms nonzero. Write $L_{J_1}(Q) = L(Q)(1+\varepsilon_Q)$ and $L_{J_2}(R) = L(R)(1+\varepsilon_R)$; their two errors have valuations at least k. Subtracting the exact ratio proves the assertion. □

For the field $\mathbb{Q}(\sqrt{2})$, every odd completion is unramified and $D = 11 - 6\sqrt{2}$ is integral. Thus these statements apply at every contact prime $p > 76176$, independently of the chosen truncation length. If M is a power of two with $M \geq 2$, the choice $J = M/2$ has relative precision at least $3M/2$, which is stronger than precision M. Arbitrarily long truncations therefore leave the fixed contact-height constant and its exceptional-prime set unchanged.

1.1 Nonvanishing at the real embeddings

Lemma 1.4. *For every integer $J \geq 1$, one has $Q_J(T) > 0$ for $0 \leq T \leq 1$.*

Proof. All c_j for $j \geq 1$ are negative. For $r \geq 1$, the recurrence telescopes to

$$\sum_{j=0}^{r} c_j = -\frac{4r-1}{2} c_r > 0.$$

The identity holds at $r = 1$ and its successive differences are c_r. Because $0 \leq T^j \leq 1$ and $c_j < 0$, it follows that $Q_J(T) \geq Q_J(1) > 0$; $J = 1$ is immediate. □

Corollary 1.5. *For $K = \mathbb{Q}(\sqrt{2})$ and $D = (3-\sqrt{2})^2$, let $Q \in 2E(K)$ be nonzero. Then $L_J(Q) \neq 0$ for every $J \geq 1$.*

Proof. The duplication identity

$$x(2R) = \left(\frac{x(R)^2 + D}{2y(R)} \right)^2$$

shows that the abscissa of a nonzero double is positive at both real embeddings. Its ordinate cannot be zero: the two nonzero ordinate-zero abscissas are $\pm(3 - \sqrt{2})$, neither a square in K since their norm is seven, while abscissa zero is excluded by the duplication formula. At either embedding the curve equation now implies $x(Q) > \sqrt{D}$. Thus $0 < Df(Q)^2 < 1$ and $z(Q) \neq 0$. Lemma 1.4 proves the assertion. □

Consequently any rational system whose logarithm denominators are evaluated at nonzero doubles may use these truncations at arbitrary finite length. Increasing the length introduces no new nonvanishing condition at those points.

1.2 A finite integer coefficient specification

For a polynomial G, write $\|G\|_1$ for the sum of the absolute values of its coefficients. In a finite free algebra over $\mathbb{Q}$ or an integer polynomial ring, choose a basis and let M_A be the matrix of multiplication by A. Use the maximum column sum of the coefficient ℓ^1 norms as its matrix norm. This norm is submultiplicative, has $\|I\| = 1$, and satisfies $\|cI\| = |c|$ for rational scalars.

Compatible weights mean nonnegative weights on base variables and positive weights on the algebra generators for which multiplication and reduction by the monic defining relations do not increase the sum of the input weights. The weighted degree of the zero element is $-\infty$; all other weighted degrees are nonnegative.

Define

$$D_J = \prod_{i=1}^{J-1} (4i+1), \qquad G_J(T) = D_J Q_J(T) \in \mathbb{Z}[T].$$

Empty products are one. Its coefficients are explicitly

$$[T^0]G_J = D_J, \qquad [T^j]G_J = 2 \left(\prod_{i=1}^{j} (4i-5) \right) \left(\prod_{i=j+1}^{J-1} (4i+1) \right) \quad (1 \leq j < J).$$

This finite product-and-sum definition specifies the ordinary integer polynomial without requiring a list of all its coefficients.

Proposition 1.6. *For $J \geq 2$, the degree of G_J is $J-1$ and*

$$D_J \leq (4J)^{J-1}, \qquad \|G_J\|_1 \leq 2J(4J)^{J-1}.$$

If $J = 2^{b-1}$, with $b \geq 2$, these imply

$$D_J < 2^{(b+1)J}, \qquad \|G_J\|_1 < 2^{(b+1)J}.$$

If A is an element of a finite free algebra with $\|M_A\| \leq 2^B$, $B \geq 0$, then

$$\|M_{G_J(A)}\| \leq \|G_J\|_1 \, 2^{(J-1)B}.$$

If the algebra has the multiplicative weighted filtration associated with compatible positive generator weights, then $\operatorname{wdeg}(G_J(A)) \leq (J-1)\max(0, \operatorname{wdeg}(A))$.

Proof. Every nonconstant coefficient is a product of $J-1$ integers of absolute value at most $4J$, times two. The same bound without two holds for the constant coefficient. Summing the coefficient bounds gives the displayed estimate. Its binary exponent when $J = 2^{b-1}$ is

$$1 + (b-1) + (J-1)(b+1) = (b+1)J - 1.$$

The operator estimate follows from the triangle inequality and submultiplicativity, and the weight estimate follows term by term. □

For a rational argument A/B, its homogeneous numerator is the explicit polynomial

$$G_J^{\mathrm{hom}}(A, B) = \sum_{j=0}^{J-1} [T^j] G_J \, A^j B^{J-1-j}.$$

Thus $Q_J(A/B) = G_J^{\mathrm{hom}}(A, B)/(D_J B^{J-1})$ on $B \neq 0$. This form is convenient for a finite algebra over an integer polynomial ring. Its degree and operator-norm estimates follow from the same proof with the larger of the two argument bounds.